Aayush Kumar

Backend & AI Engineer

+91 92043 62088 | aayush2615@gmail.com | linkedin.com/in/aayush2724 | github.com/aayush2724 | aayush-2724.vercel.app

Education

The National Institute of Engineering, Mysore Aug 2024 – May 2028
B.E. in Computer Science and Engineering (AI & ML) | CGPA: 9.3/10 *Mysuru, India*
Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Distributed Systems, Machine Learning

Experience

HSH Technologies Sep 2026 – Present
Software Engineering Intern, Backend & AI | GitHub | Live Demo *Remote*

- Built **Translog**, an LLM-assisted air-freight quotation desk in **Python**, live for **2 client mailboxes**: turns Gmail enquiries into schema-validated shipments, fetches live rates, and holds every outgoing reply for human approval.
- Designed the rate-search pipeline on **Redis + RQ**: idempotency-keyed jobs so a retry joins the existing job instead of duplicating browser work, per-mailbox state and failure isolation, a worker lock with heartbeat, and automatic requeue of failed jobs.
- Built a hand-labelled **LLM evaluation harness** over **45 real enquiries** (11 fields each); prompt and schema iterations raised field accuracy from **87.1% to 89.9%** with zero fully-wrong cases.
- Automated **WebCargo** lookups with a **Playwright** browser worker (signed-in session keep-alive, IATA resolution, goods-type rules); eligibility filtering records a reason for every excluded rate and selects the fastest eligible one. Backed by **2,400+** credential-free tests under `mypy --strict`.

OWASP NIE Student Chapter Aug 2026 – Present
President *Mysuru, India*

- Lead a **29-member core team** running web-security workshops, Burp Suite labs and secure-coding hacknights; **4** flagship events have drawn **200+** participants.
- Built and maintain the chapter website (Next.js 16, Sanity CMS synced to PostgreSQL via Drizzle, Redis read-through cache) with an event-registration API that rate-limits by IP in Redis and enforces one registration per email and USN.

Projects

Auralis – AI Sales Intelligence Platform Jul 2026
Python, FastAPI, LangGraph, LangChain, RAG, FAISS, Transformers, PostgreSQL, Redis, Docker GitHub | Live

- Built a stateful **LangGraph** agent that tags each message with **7 objection classes**, **5 personas** and sentiment via zero-shot NLI, then routes to **6 response strategies**.
- Implemented a **RAG** pipeline: PDF/CSV/Markdown ingestion, recursive chunking, Gemini embeddings and **FAISS** similarity search, with source citations injected into every reply and a 3-trigger human handoff (explicit request, low confidence, negative sentiment).
- Served it through a **FastAPI** backend with JWT auth, a **WebSocket** chat endpoint, Prometheus metrics, **PostgreSQL** analytics and **Redis** session/A-B state; containerized with **Docker**, **220+** tests run in GitHub Actions CI.

AlgoVision – Algorithm Learning Engine Jun 2026 – Oct 2026
Python, FastAPI, JavaScript (ES modules), Three.js, GSAP, Pyodide/WebAssembly, Groq, Docker GitHub | Live

- Built a **FastAPI** tracing engine with **474 live algorithm tracers** that steps through the student's own input, with a scrubber, operation counters and a mode that races two algorithms.
- Added an in-browser Python dry run on **Pyodide** (WebAssembly), an LLM narrator microservice (**Groq**) with template fallbacks, and **23** offline emulators for backend-free use.
- Enforced one hash-based CSP and security-header set across Vercel, nginx and FastAPI via tests; **860+** pytest cases cover tracers, API and headers; deployed on Vercel with `/api` rewrites to Render.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Server-Sent Events, JWT, OAuth 2.0
AI / LLM: LangChain, LangGraph, RAG, Embeddings, FAISS, Hugging Face Transformers, Prompt Engineering, LLM Evaluation
Data & Queues: PostgreSQL, Redis, MongoDB, Firebase, RQ
Infra & Tools: Docker, Linux, Git, GitHub Actions (CI/CD), Playwright, Postman, Railway, Render, Vercel
Frontend: React, TypeScript, Tailwind CSS, Three.js, GSAP

Achievements

- **3rd Place**, ThinkRoot x Vortex'26, NIT Trichy (100+ teams, 24-hr build) | **3x Hackathon Finalist** incl. InnovaHack
- Solved **770+ DSA problems** on LeetCode.